

Research Notes on Speech recognition

Speech recognition

>> Section 1

Conferences Regular conferences include SpeechTEK and SpeechTEK Europe, ICASSP, Interspeech/Eurospeech, and the IEEE ASRU. Conferences in the field of natural language processing, such as ACL, NAACL, EMNLP, and HLT, include papers on speech processing.

>> Section 2

Pre-1970 1952 – Bell Labs researchers, Stephen Balashek, R. Biddulph, and K. H. Davis, built Audrey for single-speaker digit recognition. Their system located the formants in the power spectrum of each utterance. 1960 – Gunnar Fant developed and published the source-filter model of speech production. 1962 – IBM's 16-word "Shoebox" machine's speech recognition debuted at the 1962 World's Fair. 1966 – Linear predictive coding, a speech coding method, was proposed by Fumitada Itakura of Nagoya University and Shuzo Saito of Nippon Telegraph and Telephone. 1969 – Funding at Bell Labs came to a halt for several years after the company's head engineer, John R. Pierce, wrote an open letter criticizing speech recognition research. This defunding lasted until Pierce retired and James L. Flanagan took over. Raj Reddy was the first person to work on continuous speech recognition, as a graduate student at Stanford University in the late 1960s. Previous systems required users to pause after each word. Reddy's system issued spoken commands for playing chess. Around this time, Soviet researchers invented the dynamic time warping (DTW) algorithm and used it to create a recognizer capable of operating on a 200-word vocabulary. DTW processed speech by dividing it into short frames (e.g. 10 ms segments) and treating each frame as a unit. Speaker independence, however, remained unsolved.

>> Section 3

Acoustic signals are structured into a hierarchy of units, e.g. phonemes, words, phrases, and sentences; Each level provides additional constraints; e.g., known word pronunciations or legal word sequences, which can compensate for errors or uncertainties at a lower level; This hierarchy of constraints improves accuracy. By combining decisions probabilistically at all lower levels, and making ultimate decisions only at the highest level, speech recognition is broken into several phases. Computationally, it is a problem in which a sound pattern has to be recognized or classified into a category that represents a meaning to a human. Every acoustic signal can be broken into smaller sub-signals. As the more complex sound signal is divided, different levels are created, where at the top level are complex sounds made

of simpler sounds on the lower level, etc. At the lowest level, simple and more probabilistic rules apply. These sounds are put together into more complex sounds on upper level, a new set of more deterministic rules predicts what the complex sound represents. The upper level of a deterministic rule should figure out the meaning of complex expressions. In order to expand our knowledge about speech recognition, we need to take into consideration neural networks. Neural network approaches use the following steps:

>> Section 4

where s is the number of substitutions, d is the number of deletions, i is the number of insertions, and n is the number of word references. While computing, the word recognition rate (WRR) is used. The formula is:

>> Section 5

Error rates increase as the vocabulary size grows: e.g. the 10 digits "zero" to "nine" can be recognized essentially perfectly, but vocabulary sizes of 200, 5000, or 100000 may have error rates of 3%, 7%, or 45% respectively. Vocabulary is hard to recognize if it contains confusing letters: e.g. the 26 letters of the English alphabet are difficult to discriminate because they are confusing words (most notoriously, the E-set: "B, C, D, E, G, P, T, V, Z (when "Z" is pronounced "zee" rather than "zed", depending on region); an 8% error rate is considered good for this vocabulary. Speaker dependence vs. independence: A speaker-dependent system is intended for use by a single speaker. A speaker-independent system is intended for use by any speaker (more difficult). Isolated, discontinuous or continuous speech With isolated speech, single words are used, which is easier to recognize. With discontinuous speech, full sentences separated by silence are used. The silence is easier to recognize similar to isolated speech. With continuous speech naturally spoken sentences are used, which are harder to recognize. Task and language constraints can inform the recognition The requesting application may dismiss the hypothesis "The apple is red." Constraints may be semantic; rejecting "The apple is angry." Syntactic; rejecting "Red is apple the." Constraints are often represented by grammar. Read vs. spontaneous speech When a person reads it's usually in a context that has been previously prepared. When a person speaks spontaneously, recognition must deal with disfluencies such as "uh" and "um", false starts, incomplete sentences, stuttering, coughing, and laughter) and limited vocabulary. Adverse conditions environmental noise (e.g., in a car or factory). Acoustic distortions (e.g. echoes, room acoustics) Speech recognition is a multi-level pattern recognition task.

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Further reading Cole, Ronald; Mariani, Joseph; Uszkoreit, Hans; Varile, Giovanni Battista; Zaenen, Annie; Zampolli, Zue, Victor, eds. (1997). Survey of the state of the art in human language technology. Cambridge Studies in Natural Language Processing. Vol. XII–XIII. Cambridge University Press. ISBN 978-0-521-59277-2. Junqua, J.-C.; Haton, J.-P. (1995). Robustness in Automatic Speech Recognition: Fundamentals and Applications. Kluwer Academic Publishers. ISBN 978-0-7923-9646-8. Karat, Clare-Marie; Vergo, John; Nahamoo, David (2007). "Conversational Interface Technologies". In Sears, Andrew; Jacko, Julie A. (eds.). The Human-Computer Interaction Handbook: Fundamentals, Evolving Technologies, and Emerging Applications (Human Factors and Ergonomics). Lawrence Erlbaum Associates Inc. ISBN 978-0-8058-5870-9. Pieraccini, Roberto (2012). The Voice in the Machine. Building Computers That Understand Speech. The MIT Press. ISBN 978-0-262-01685-8. Pirani, Giancarlo, ed. (2013). Advanced algorithms and architectures for speech understanding. Springer Science & Business Media. ISBN 978-3-642-84341-9. Signer, Beat; Hoste, Lode (December 2013). "SpeeG2: A Speech- and Gesture-based Interface for Efficient Controller-free Text Entry". Proceedings of ICMI 2013. 15th International Conference on Multimodal Interaction. Sydney, Australia. Woelfel, Matthias; McDonough, John (26 May 2009). Distant Speech Recognition. Wiley. ISBN 978-0-470-51704-8.

>> Section 7

Aerospace, e.g., NASA's Mars Polar Lander used speech recognition technology from Sensory, Inc. in the Mars microphone on the lander Automatic subtitling with speech recognition Automatic emotion recognition Automatic shot listing in audiovisual production Automatic translation E-discovery Hands-free computing Home automation Interactive voice response Mobile telephony, including mobile email Multimodal interaction Real-time captioning Robotics Security, including usage with other biometric scanners for multi-factor authentication Speech to text Telematics, e.g., vehicle navigation systems Transcription Video games like Tom Clancy's EndWar, Bow-wow Battle, and Lifeline Virtual assistant such as Siri